DIPLOMA IN INFORMATION TECHNOLOGY

ASSIGNMNET 1

Subject: **MOBILE APPLICATION**
Subject code: **DIT2223**

DEADLINE: Week 8 Monday

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Student ID: **24052573**

SDS ACADEMIC INTEGRITY STATEMENT

Sunway Diploma Studies is committed to the principles of academic integrity. Academic integrity means placing five fundamental values: **honesty, trust, fairness, respect, and responsibility** into practice. It is being honest in the academic work you do at the programme/college, being fair to others, and taking responsibility for learning, and acting in an ethical manner in all your academic endeavours. We believe that these five values are truly foundational to the programme.

I hereby declare that:

1. I fully understand and will uphold the academic integrity of Sunway Diploma Studies (SDS).
2. I confirm that the work hereby submitted is my own original work and where other people's work has been used this has been fully acknowledged.
3. I am aware of the importance of conducting exams with integrity and fairness, and I hereby confirm that I will comply with the requirements.
4. I am aware that non-compliance with these instructions and unfair conduct constitute a disciplinary offense, and actions will be taken including but not limited to being expelled.

<Please sign here>

…….

Name: Koh Pei Yang

Student ID: 24052573

<Please put the date here>

……..

Date: 9/6/2025

Introduction

This report introduces the functionality and conceptual logic of the 'Pokémon Clicker' application, explaining how it is designed to be used and how users can interact with it.

Application Name: **Pokémon Clicker**
Type: **Game**
Support Device: **Mobile**
Game Style: **Clicker Game**
Game Topic: **Eevee (**a type of Pokémon in **Pokémon)**

1. **What is Pokémon clicker?**
   Pokémon clicker game is a type of incremental or idle game themed around the Pokémon franchise. User needs to click or tap the screen repeatedly to evolve the Pokémon until the final evolution of 'Eevee'.

2. **Why I build this application?**
   As a Pokémon's fan, I like the design of different type of Pokémon, especially 'Sylveon'. So, I decided to build a Pokémon-themed game, to relive my childhood memories. Pokémon games are usually simple and relaxing, making all ages of player enjoy the gameplay. That's why I choose a clicker style game where players tap the screen to level up.

3. **Why did I use a 2D pixel image?**
   I used 2D pixel format in the game. This making the whole game become retro style, also to respect the original Pokémon's gameplay.

4. **Why I used 'Eevee' as my topic?**
   'Eevee' have different type of evolution in game, it have across 8 evolutionary in the game. I want to challenge different scene settings, music and animation of 8 evolution.

Learn From Project:

1. **Display a GIF image in android studio:**
   In my whole activity, I used GIF image to present my animation, although it may be simple and crude, but I learn to combine a timer to make my Pokémon evolve.

2. **Display a pop out window:**
   I used a pop out window in my setting window, confirm reset setting and complete window setting. I also setting the size of pop out window and make it transparent.

3. Reset a progress and activity

I set a reset value to return the java class and launch the setting's activity, when it completes the game, there will be asked user to confirm reset the progress to continue the gameplay.

4. Randomize evolution of Eevee

I set a random class to randomize the 8 evolutions, which means that user may evolve differently evolution or the same evolution. I think this is surprised for user which have 1/8 chances to get their favourite evolution.

5. Setting a music control

This can help me to stop the music while changed to other activity. It also helps to pause the music while I press the setting buttons.

Own Performance Evaluation:

1. Good visual effect & music

Each activity page has its unique background image and music, which enhances the immersive and enjoyable for user.

2. Lack of functionality in setting

The setting menu is missing a music or sound control function, which is limited to user's ability to customize their experience.

Challenge Faces:

1. Implement the gif image

It is hard to import the glide library in android studio, because I have no idea to locate a build.gradle.kts. After I found the tutorial in website, then I solve it. But some of gif image source is not suitable for my design and hard to find in website, especially the gif's animation time and size.

2. Multiple activity process at the same time

When I reset the progress in Complete_Window, the old activity still running in backside program. This is a valid concern especially repeatedly opening Complete_Window and transitioning to Eevee activity, affects the gif image's animation load slower and the program overload.

3. Design an icon by using the tools in activity

It is hard for me to design a pixel icon in the Android Studio's activity. It involves careful manipulation of individual pixel to create visually appealing and detailed image, that why I am using a Canva to design a picture and paste it as image in Android Studio. Also, my setting button lost the icon of music volume control is because of this.

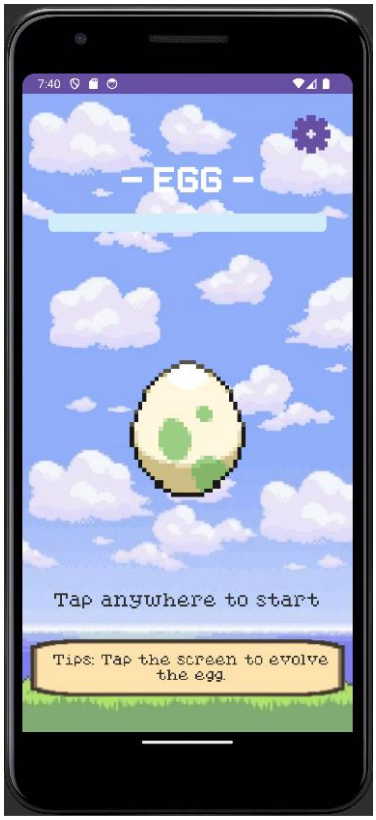
Screenshot

Loading Page & Egg Page

The gif image logo will pop out like 5 second, then change to Egg's activity. When user tap the screen, it will show the progress in progress bar on top, like picture 3.

Every top of progress bar has the Pokémon's name, here show you the name is 'Egg'. The bottom also shows some tips to help user start the game.

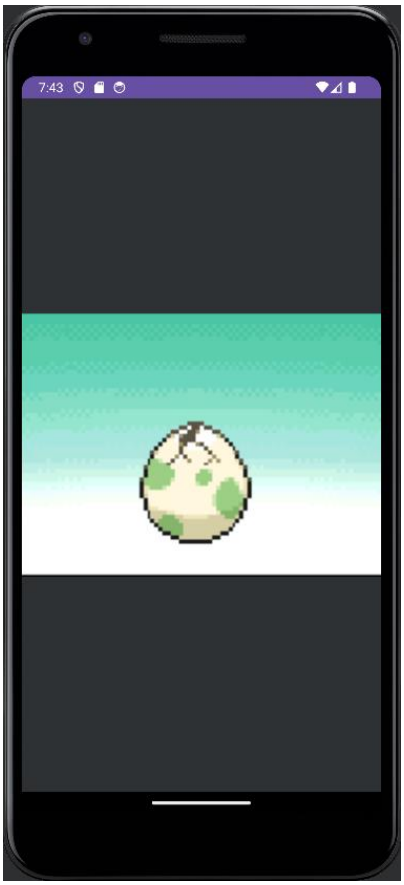
Setting Page & Confirm Reset Page

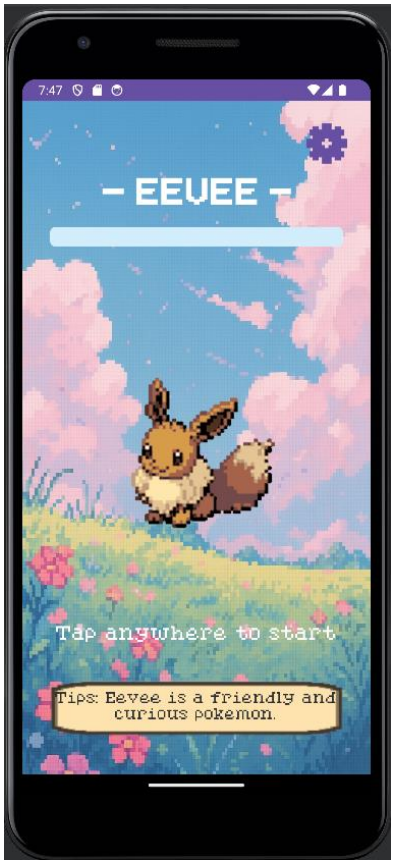


When user touch the setting button in top right corner, there will pop out the setting menu. Here show you the reset progress button and the Eevee evolutions image.

In picture 2, When user click the reset button, it will pop out the reset confirm window to ask user for reset. If yes, the progress reset (picture 3), if no, the window close. (return to picture 1)

Egg_Evolve Page & Eevee Page

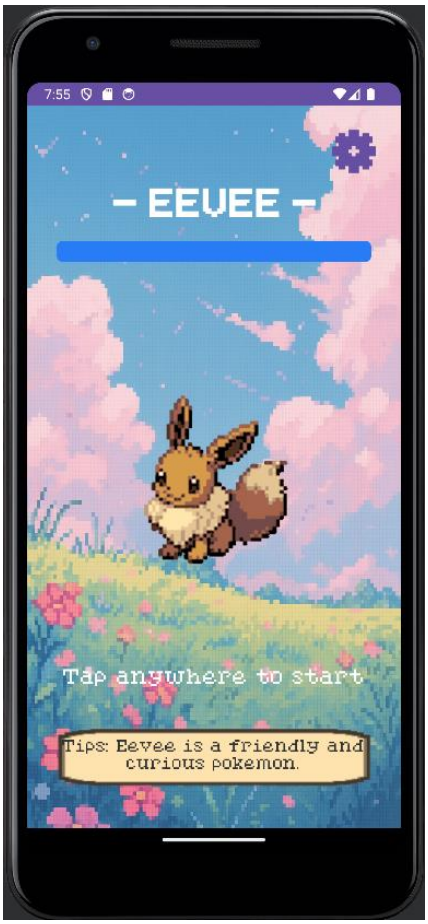


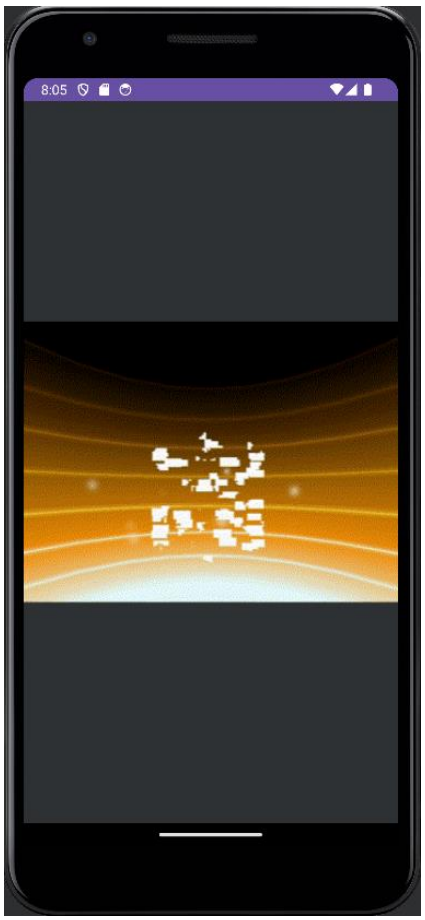


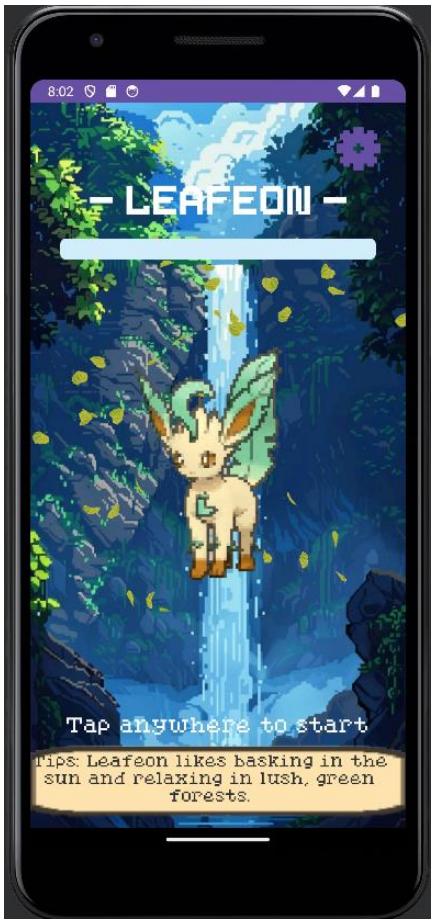
When the progress bar is full (10 Clicks), the egg will turn to Egg_Evolve's page (Picture 2).

After the Egg_Evolve's animation, start the Eevee's page (Picture 3).

Eevee_Evolve Page & Leafeon Page



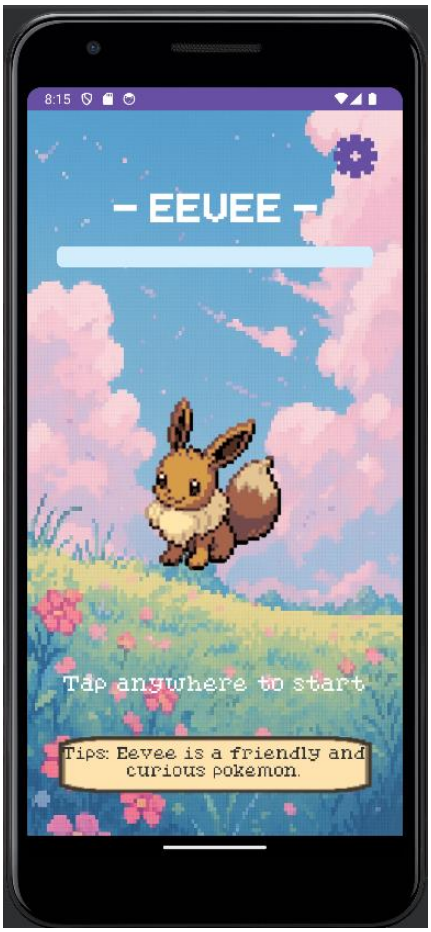




After 15 Clicks, the Eevee page turn to Eevee's_Evolve page (Picture 2).

After 5.5 seconds, there will turn to Leafeon's page (Picture 3).

Complete_Window & Close Application

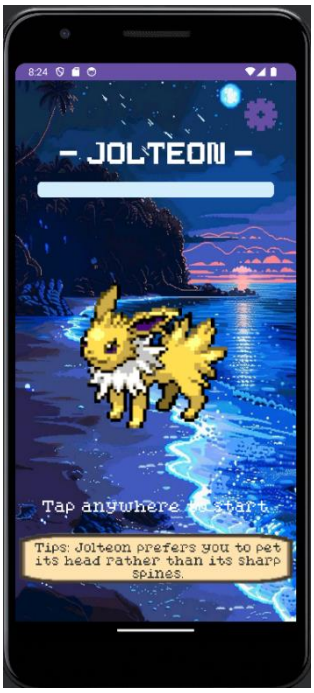


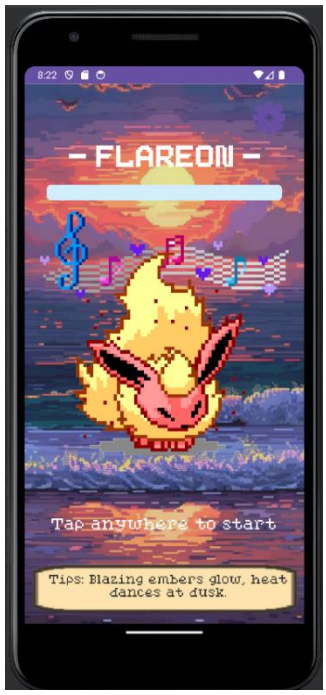
After the progress bar full (10 Clicks), start the Complete_Window's page.

If user choose yes, turn to Eevee page (Picture 2).

If no, close the application (Picture 3).

All Eevee_Evole's page





Here is all the evolution of Eevee also have its unique background music.

Marking Rubric for Assignment

The assignment will be marked based on the following criteria. Include this sheet in the submission.

Criteria	Very Poor	Poor	Good	Expert	Allocated Marks
Program Execution	(0 – 4 marks) Weak Program execution with major flaws	(5 - 9 marks) Average Program execution with some flaws	(10 - 14 marks) Good Program execution with minor flaws	(16 - 20 marks) Excellent Program execution	18
Program Correctness (compliant with the specifications stated in assignment)	(0 – 4 marks) Little coverage of assignments requirements with weak implementation	(5 – 9 marks) Comply some of the assignment requirements with average or below average implementation	(10 – 14 marks) Comply with the most of the assignment requirements with good implementation	(16 - 20 marks) Fully comply with the assignment requirements with excellent implementation	18
Creativity & Innovative	(0 – 4 mark) Weak creativity product produced	(5 – 9 marks) Average product with average creativity	(10 - 14 marks) Good product with good creativity idea	(16 – 20 marks) Excellent product which full with creativity and fresh idea	18
Output & Layout	(0 – 4 mark) Weak/no output & layout produced	(5 – 9 marks) Poor/Average output & layout produced	(10 - 14 marks) Good output & layout produced	(16 – 20 marks) Excellent output & layout produced	18
Reflection Report	(0 – 1 mark) Weak report provided with major omission	(2 marks) Poor/Average report provided but some aspects are not elaborated in details	(3 marks) Good report provided with minor omission in certain aspects	(4 – 5 marks) Excellent reflection report provided	4
Presentation	(0 – 7 mark) Weak and unable to demonstration product clearly	(8 marks) Poor/Average in overall presentation	(9 – 11 marks) Good in overall presentation	(12 – 15 marks) Excellent in overall presentation	12

Total: 88 / 100